

Stat 435 Lecture Notes 5a

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Logistic regression: diagnostics

Recap on model and assumptions

- Y is a Bernoulli random variable and $E(Y = 1|X) = p(X)$
- Population model:

$$\text{logit}(p(X)) = \log\left(\frac{p(X)}{1-p(X)}\right) = \beta_0 + \beta_1 X \quad (1)$$

or equivalently

$$p(X) = \frac{\exp(\beta_0 + \beta_1 X)}{1 + \exp(\beta_0 + \beta_1 X)} = \frac{1}{1 + \exp(-\beta_0 - \beta_1 X)} \quad (2)$$

- Sample version of the model:

$$\text{logit}(p(x_i)) = \beta_0 + \beta_1 x_i$$

for the i th observation (x_i, y_i) of (X, Y) .

Recap on assumptions

The assumptions of the model are:

- Each y_i is Bernoulli random variable with expectation $\mu_i = p(x_i)$
- $\text{logit}(p(X))$ is a linear function of X , i.e., $\text{logit}(p(x_i))$ is a linear function of x_i for all $1 \leq i \leq n$
- The pairs $\{(x_i, y_i)\}_{i=1}^n$ are independent

Fitted values

Once an estimate $(\hat{\beta}_0, \hat{\beta}_1)$ of (β_0, β_1) is obtained, we have the estimated model

$$\text{logit}(\hat{p}(x_i)) = \hat{\beta}_0 + \hat{\beta}_1 x_i,$$

where the mean or probability of success μ_i for each y_i is estimated by

$$\hat{\mu}_i = \hat{p}(x_i) = \frac{1}{1 + \exp(-\hat{\beta}_0 - \hat{\beta}_1 x_i)}$$

Ordinary residuals

Formula:

$$e_i = \begin{cases} 1 - \hat{\mu}_i & \text{if } y_i = 1 \\ -\hat{\mu}_i & \text{if } y_i = 0 \end{cases}$$

Note: the distribution of $\{e_i\}_{i=1}^n$ is unknown even if the fitted model is correct

Pearson residuals

- Formula:

$$r_i = \frac{y_i - \hat{\mu}_i}{\sqrt{\hat{\mu}_i(1 - \hat{\mu}_i)}}.$$

Note that $\sqrt{\hat{\mu}_i(1 - \hat{\mu}_i)}$ can be thought of as an estimate of the standard deviation of $y_i - \hat{\mu}_i$

- Studentized Pearson residuals:

$$\tilde{r}_i = \frac{r_i}{\sqrt{1 - h_{ii}}},$$

where h_{ii} is the i -th diagonal entry of

$$H = W^{1/2} X (X' W X)^{-1} X' W^{1/2}$$

and W is the $n \times n$ diagonal matrix with entries $\hat{\mu}_i(1 - \hat{\mu}_i)$ and X is the $n \times p$ design matrix

Deviance residuals

- The “probability mass function (PMF)” of y_i , or the likelihood function of y_i , is

$$f_i(y_i) = p_i^{y_i} (1 - p_i)^{1 - y_i}$$

- Let the estimated log-likelihood function of y_i be denoted by

$$\ln \hat{f}_i(y_i) = y_i \ln \hat{\mu}_i + (1 - y_i) \ln (1 - \hat{\mu}_i)$$

- The *model deviance* is

$$D = -2 \sum_{i=1}^n \ln \hat{f}_i(y_i) = \sum_{i=1}^n d_i^2, \quad (3)$$

where the *i*th *deviance residual* is

$$d_i = \text{sgn}(y_i - \hat{\mu}_i) \sqrt{-2 \ln \hat{f}_i(y_i)}$$

Deviance residuals

- The model deviance D is equal to

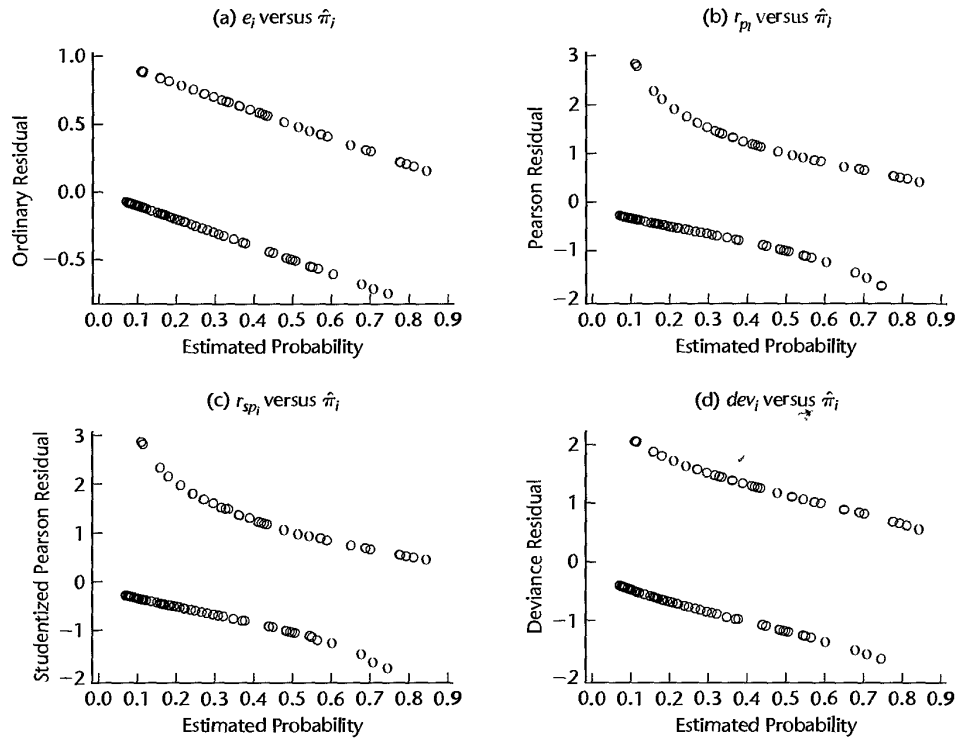
$$-2 \times [\ln L(\text{Reduced model}) - \ln L(\text{Full model})],$$

where the “Reduced model” is the model under consideration and the “Full model” is the full nonparametric model of $E(y_i) = \mu_i = \Pr(Y = 1)$

- Under appropriate conditions, D approximately has a chi-square distribution when the Reduced model is correct. D itself can be used to check if the postulated linear model is appropriate.

Checking model fit

FIGURE 14.12 Selected Residuals Plotted against Predicted Mean Response—Disease Outbreak Example.



Checking model fit

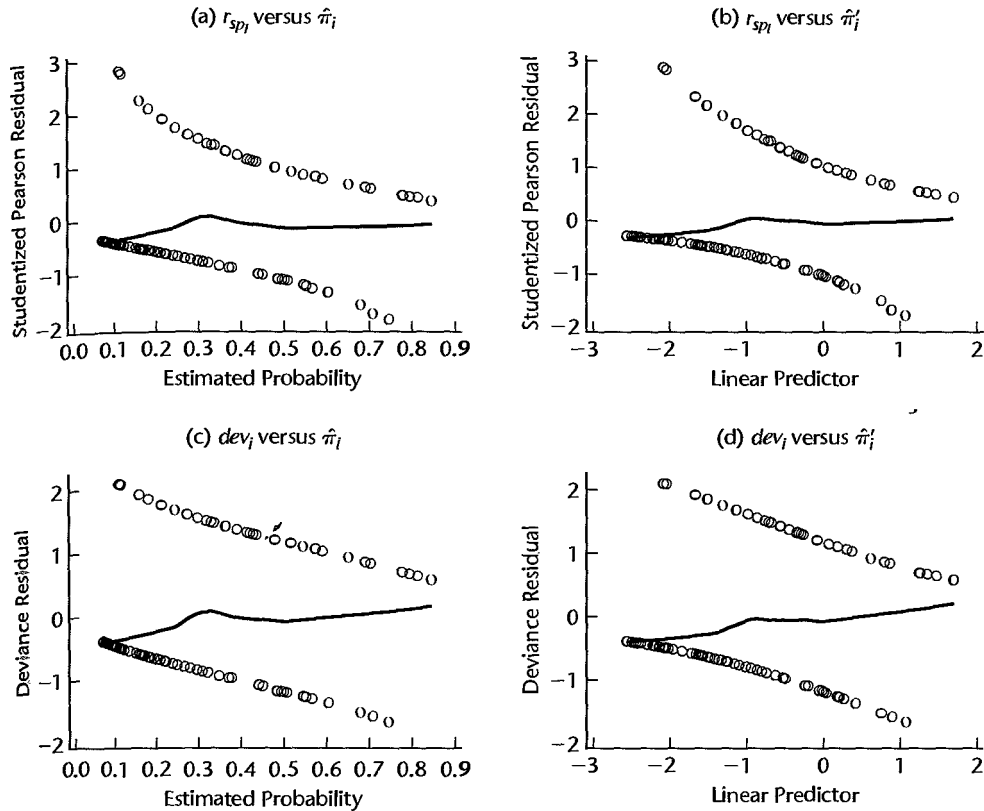
- If the logistic model is correct, then $E(y_i) = \mu_i$ and asymptotically

$$E(y_i - \hat{\mu}_i) = 0$$

- Thus, if the model is correct, a lowess smooth fit of the plot of the residuals against the estimated probabilities (or against the linear predictor $\hat{\eta}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$) should result approximately in a horizontal line with zero intercept
- Any significant departure from this suggests that the model may be inadequate. In practice, either the ordinary residuals, the Pearson residuals, or the studentized residuals can be used for this purpose

Checking model fit

FIGURE 14.13 Residual Plots with Lowess Smooth—Disease Outbreak Example.



Checking influential observations

- Recall *Pearson chi-square goodness of fit statistic*, in the setting of logistic regression, as

$$\xi^2 = \sum_{i=1}^n \frac{(y_i - \hat{\mu}_i)^2}{\hat{\mu}_i(1 - \hat{\mu}_i)} = \sum_{i=1}^n r_i^2$$

- Denote by $\xi_{(i)}^2$ the value of the Pearson chi-square goodness of fit statistic obtained when observation i is deleted

- Denote by $D_{(i)}$ the value of the model deviance obtained when observation i is deleted

Checking influential observations

- Define the “delta chi-square statistic” as

$$\Delta\xi_{(i)}^2 = \xi^2 - \xi_{(i)}^2$$

and “delta deviance statistic” as

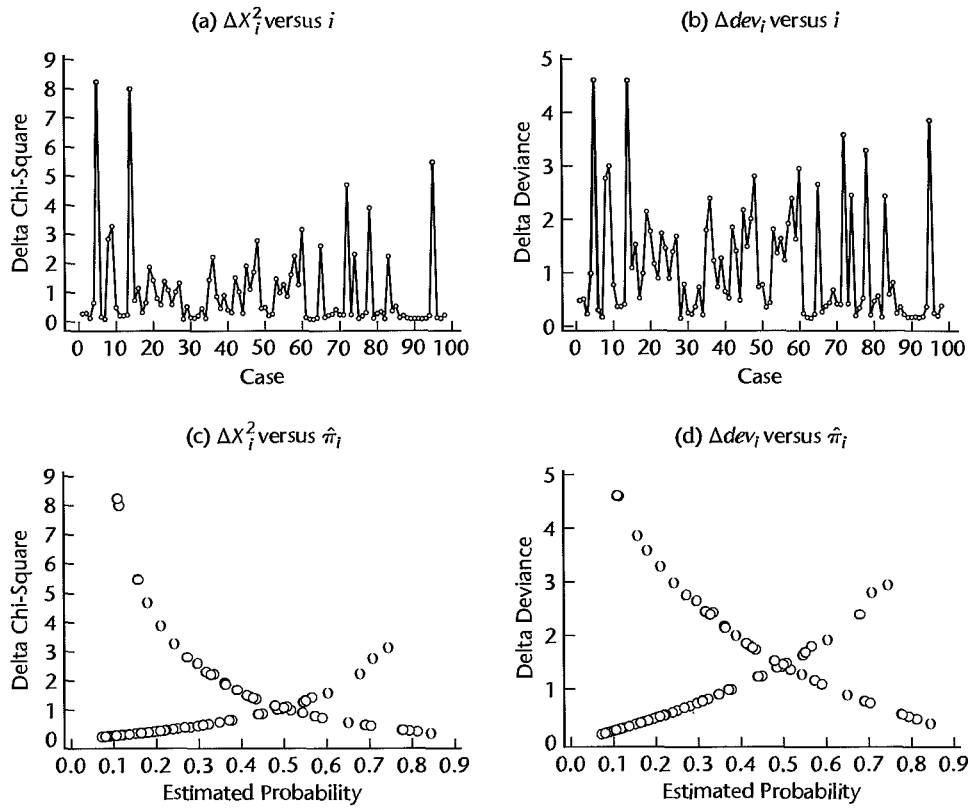
$$\Delta D_{(i)} = D - D_{(i)}.$$

These quantities measure the influence of the i th observation on model fitting

- Usually, these statistics are plotted against the observation number i , against $\hat{\mu}_i$, or against $\hat{\eta}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$. Extreme values appear as spikes when plotted against i , or as outliers in the upper corners of the plot when plotted against $\hat{\mu}_i$ or $\hat{\eta}_i$

Checking influential observations

FIGURE 14.15 Delta Chi-Square and Delta Deviance Plots—Disease Outbreak Example.



Checking influential observations

Cook's distance is defined as

$$\delta_i = \frac{r_i^2 h_{ii}}{p(1 - h_{ii})^2}$$

where

- h_{ii} is the i -th diagonal entry of

$$H = W^{1/2} X (X' W X)^{-1} X' W^{1/2}$$

- W is the $n \times n$ diagonal matrix with entries $\hat{\mu}_i (1 - \hat{\mu}_i)$ and X is the $n \times p$ design matrix.

License and session Information

License

```
> sessionInfo()
R version 3.5.0 (2018-04-23)
Platform: x86_64-w64-mingw32/x64 (64-bit)
Running under: Windows 10 x64 (build 19041)

Matrix products: default

locale:
[1] LC_COLLATE=English_United States.1252
[2] LC_CTYPE=English_United States.1252
[3] LC_MONETARY=English_United States.1252
[4] LC_NUMERIC=C
[5] LC_TIME=English_United States.1252

attached base packages:
[1] stats      graphics  grDevices  utils      datasets  methods
[7] base

other attached packages:
[1] knitr_1.21

loaded via a namespace (and not attached):
[1] compiler_3.5.0  magrittr_1.5    tools_3.5.0
[4] htmltools_0.3.6 yaml_2.2.0      Rcpp_1.0.0
[7] stringi_1.2.4   rmarkdown_1.11  stringr_1.3.1
[10] xfun_0.4        digest_0.6.18   evaluate_0.12
```